

What we learned from 1,200 hours of mentor pairings

An honest look at what 1,200 hours of one-on-one mentorship taught us about matching, retention, and the small rituals that decide whether a pairing sticks.

S. Li · 12 min read · December 2025

When we started Growing Up with Robotics, we thought the hard part was going to be the curriculum. Which kit to use. How to sequence Scratch before Python before C++. Whether a ten-year-old could really handle a PID controller if we just explained it the right way. We spent months on those questions. We were wrong about which ones mattered.

The hard part, it turned out, was the human pairing. The thirty seconds at the start of a session where a mentor either remembered what their student had been stuck on last week, or didn't. The decision to spend an entire hour debugging a single motor instead of moving on to the lesson plan. The small, almost invisible rituals that decided whether a student showed up the next Saturday or quietly stopped coming.

This is what 1,200 hours of mentor pairings taught us. I want to be honest about it, because most of what gets written about STEM outreach is celebratory, and most of what we did in our first year didn't work. The things that eventually did work were smaller and stranger than I would have predicted.

The trap of the impressive demo

Our earliest sessions were built around what I now think of as the impressive demo. A mentor would show up with a fully assembled robot, run it through its paces, and then walk a group of students through "building" something similar. The students would leave excited. Parents would send thank-you notes. We would tell ourselves we had made an impact.

Three weeks later, when we asked those same students what they had built, most of them couldn't remember. Not the details, not the concept, not even what the robot was supposed to do. They remembered that it was cool. They remembered the mentor's name,

sometimes. They did not remember anything they could do themselves now that they couldn't do before.

This is the trap. Impressive demos look like impact because everyone in the room is smiling. But a smiling student is not a learning student, and the gap between those two things is where most outreach programs quietly fail. We were treating mentorship like a performance, where the mentor was the performer and the student was the audience. Real mentorship is closer to the opposite. The student is doing the work. The mentor is mostly invisible, occasionally indispensable, and rarely the most interesting person in the room.

The first thing we changed was the rule that a mentor was not allowed to touch the robot. Not to fix it, not to demonstrate, not even to show what a correct assembly looked like. If the student needed to see something, the mentor could draw it on paper or point at a diagram. The student had to put the screw in. The student had to feel the click of the LEGO Technic pin going into the wrong hole, and then the right one. This change alone, more than any curriculum revision, was responsible for our retention improving.

Meeting students where they actually are

There is a phrase that gets used in education a lot, "meeting students where they are," and most of the time when people say it they mean something fairly soft, like being patient or being kind. What we found is that it means something much more specific and much more uncomfortable. It means accepting that the student in front of you is not the student you wish you had, and building from there.

The student in front of you might be eight years old and convinced she is bad at math because someone at school told her so. The student in front of you might be twelve and able to recite the entire Wikipedia article on quaternions but unable to identify which side of a battery is positive. The student in front of you might be a fourteen-year-old who has built three Arduino projects already and is bored to tears by your introductory lesson on what a resistor does. None of these students need the same hour from you. The pre-written lesson plan, no matter how good it is, is going to fail at least two of them.

We trained our mentors to spend the first ten minutes of every pairing on what we started calling a tare. The word comes from weighing things. You put the empty container on the scale and zero it out, so that what you measure next is just the thing you actually care about. A mentorship tare is the same idea. What does this kid already know? What do they think they know that they don't? What are they secretly afraid of? You can't teach them anything useful until you've measured the baseline, and you can't measure the baseline by

asking them what they know, because most kids will either lie out of pride or lie out of shame. You have to give them something to do and watch.

A good tare for a beginner is "build me anything that moves." Not anything specific. Anything. A bad mentor will look at this and see a vague prompt. A good mentor will watch for forty things in the first three minutes. Does the student reach for the wheels first or the motors? Do they try to attach the battery before they have a frame? Do they look up at the mentor for approval after every step, or do they just keep going? Each of these tells you something about what to do for the next fifty minutes, and none of it is on the lesson plan.

The case for the smallest possible improvement

Once you know where the student actually is, the next mistake is to try to move them too far in one session. This was our second big lesson, and it took us most of a year to internalize it.

The instinct, especially for technically strong mentors, is to teach the most interesting thing they can. If a student has just figured out how to make a motor spin, the temptation is to immediately introduce variable speed, then direction, then a sensor input, then a feedback loop, because each of these is genuinely cool and each one builds on the last. By the end of the hour the student has been exposed to four new concepts. By the start of the next session they remember zero of them.

What works better, almost embarrassingly so, is to pick one small thing and make sure it sticks. Not "let's learn about sensors today." Something like "by the end of this hour, you will know how to make the robot stop when it sees the wall." That's it. That's the whole goal. If you accomplish it in fifteen minutes, you spend the remaining forty-five minutes letting the student modify the threshold, break it in interesting ways, predict what will happen if the wall is further away, and explain it back to you in their own words. The concept gets pressed into them through repetition and play, rather than introduced and then immediately abandoned for the next concept.

This is hard for mentors. It feels like you are wasting the student's time. It feels like you should be teaching more. The thing you have to keep reminding yourself is that the student is not there to be impressed by how much you can teach. The student is there to leave the session knowing one thing they did not know when they came in, and ideally to be able to demonstrate it to a parent or a friend that night at dinner. One thing, demonstrated well, is worth more than five things mentioned. We have tracked this. The students who can teach their parents what they learned at the end of a session are dramatically more likely to be there the next week.

Tracking improvement is harder than it sounds

The thing I most wish we had done from the beginning was track student progress in a way that was actually useful. We did track things. We tracked attendance, we tracked sessions completed, we tracked which kits each student had touched. None of these turned out to mean what we thought they meant.

A student can attend twenty sessions and learn very little if every session is functionally a repeat of the last. A student can complete a kit by following the instructions step by step and still not understand any of the underlying ideas. The metrics we were collecting told us about activity, not about learning, and the gap between those two is the same gap that swallows most educational interventions. You can be very busy and produce nothing.

What we eventually started tracking, with mixed success, is something we call the unprompted demo. At the end of every session, the mentor asks the student to show them something they learned, without prompting them on what to choose. If the student can pick something, set it up, and explain it, that counts. If they freeze, or look at the mentor for what to demonstrate, or describe it in vague terms ("we learned about motors"), that does not count. We log it as a yes or a no. Over time, for each student, you get a series of yeses and nos that actually tells you something. A student who has six consecutive yeses is genuinely accumulating skills. A student who has six consecutive nos is showing up but not learning, and that is a problem to be solved at the program level, not the student level.

The unprompted demo is harder to game than a quiz, because the student is choosing what to demonstrate. It is more honest than self-reported confidence, because confidence is not the same as competence and any teacher who has worked with kids knows that the loudest student in the room is often not the one who has actually learned the most. It is more useful than a grade, because it tells you specifically what they can do, which is the thing you wanted to teach them in the first place.

We also started keeping what we called drift notes. After every session, the mentor wrote one or two sentences about what the student had been working on, what was hard, what they seemed to enjoy, and what they wanted to try next time. The notes were for the next mentor, in case of a swap, but they were also a forcing function for the current mentor. You cannot write a useful drift note about a session in which you didn't actually pay attention to the student. The act of writing it three minutes after the session ends forces you to remember whether you were teaching a person or just running a lesson plan.

What the small rituals add up to

If I had to summarize what 1,200 hours of pairings actually taught us, it would be that mentorship is not the transfer of expertise from a person who has it to a person who doesn't. That is what it looks like from the outside, and it is what most program designs implicitly assume, but it is not what works. Mentorship is the construction, over time, of a relationship in which one person's small improvements become visible, repeatable, and valued.

Every piece of that sentence matters. The improvements have to be small, because big improvements are mostly a fiction we tell ourselves to feel good. They have to be visible, because invisible progress doesn't compound, doesn't get celebrated, and doesn't keep a kid coming back. They have to be repeatable, because if the student can only do the thing with the mentor sitting next to them, the mentor is the skill, not the student. And they have to be valued, by the mentor, by the student, by the people around the student, because skills that no one notices tend to atrophy quietly.

The rituals we have built up over time, the no-touch rule, the ten-minute tare, the single small goal, the unprompted demo, the drift note, are all aimed at exactly this. None of them are clever. Most of them are the kind of thing an experienced classroom teacher would recognize immediately and probably be slightly amused that we presented as discoveries. But the difference between knowing those things in the abstract and actually doing them in every single session with every single mentor is, I have come to believe, the entire game.

We are still bad at a lot of this. Some pairings still drift. Some mentors still default to the impressive demo when they are tired. Some students still show up for six weeks and then disappear and we never figure out why. But the floor is higher than it used to be, and the ceiling is higher too, and the students who stick with us long enough to teach the next cohort are the strongest argument we have that any of this is working.

If you are running a program like ours, or thinking about starting one, the only piece of advice I would feel confident giving you is this. Track the small things. Make your mentors do the small things even when the small things feel beneath the dignity of the work. Resist the urge to be impressive. The students will tell you, by their feet and by their faces, whether what you are doing is real or not. Listen to them. They are right more often than we are.

Growing Up with Robotics is a 501(c)(3) nonprofit running one-on-one and small-group robotics mentorship for students aged 8 to 16. To learn more, volunteer, or sponsor a cohort, visit our website.